

Exercise 1 – Solution Proposal

Thermodynamics of Electrochemical Cells

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Interface Theory and Computational Electrocatalysis

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Introduction

In this document, we investigate cell potentials. The questions and notation are taken from the problem sheet [1]. The formulas and Table used in this document are

Equations: Standard cell potential

And standard conditions, i. e. pressure is $p^\ominus = 1$ bar, the standard cell potential is

$$E_{\text{cell}}^\ominus = E_{\text{red, cathode}}^\ominus - E_{\text{red, anode}}^\ominus \quad (1)$$

Table 1: Electrochemical series

Half-cell			$E_{\text{red}}^\ominus$ (V)
$\text{Ag}^+(\text{aq}) + \text{e}^-$	$\rightarrow$	$\text{Ag}(\text{s})$	+0.80
$\text{Cu}^{2+}(\text{aq}) + 2\text{e}^-$	$\rightarrow$	$\text{Cu}(\text{s})$	+0.34
$\text{Ni}^{2+}(\text{aq}) + 2\text{e}^-$	$\rightarrow$	$\text{Ni}(\text{s})$	-0.25
$\text{Fe}^{2+}(\text{aq}) + 2\text{e}^-$	$\rightarrow$	$\text{Fe}(\text{s})$	-0.44
$\text{Zn}^{2+}(\text{aq}) + 2\text{e}^-$	$\rightarrow$	$\text{Zn}(\text{s})$	-0.76
$\text{Al}^{3+}(\text{aq}) + 3\text{e}^-$	$\rightarrow$	$\text{Al}(\text{s})$	-1.66

* Values are measured at 1 bar and 298.15 K.

Equations: Nernst equation

Let E_{cell} be the cell potential and Q be the reaction quotient of an electrochemical reaction. Then the cell potential is found using the Nernst equation:

$$E_{\text{cell}} = E_{\text{cell}}^\ominus - \frac{RT}{\nu F} \ln(Q) \quad (2)$$

where R is the ideal gas constant, T is the temperature, ν is the number of electrons transferred, and F is the Faraday constant.

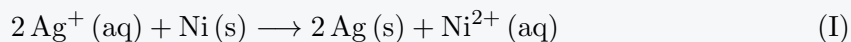
Furthermore, at equilibrium we have $Q = K$, and no potential is generated by the galvanic cell, i.e. $E_{\text{cell}} = 0$. Therefore,

$$E_{\text{cell}}^\ominus = \frac{RT}{\nu F} \ln(K) \quad \Rightarrow \quad K = \exp\left(\frac{\nu F E_{\text{cell}}^\ominus}{RT}\right) \quad (3)$$

1 The Galvanic cell

Question: 1.1 Silver and Nickel

For the following electrochemical reaction



1.1a) Write the half-cell reactions.

1.1b) Calculate the standard cell potential.

Solution: 1.1 Silver and Nickel

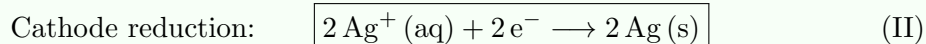
1.1a) Half-cell reactions

From Table 1, we have

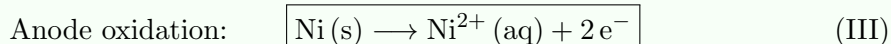
$$E_{\text{red}}^{\ominus} (\text{Ag}^+/\text{Ag}) = 0.80 \text{ V} \quad (4\text{a})$$

$$E_{\text{red}}^{\ominus} (\text{Ni}^{2+}/\text{Ni}) = -0.25 \text{ V} \quad (4\text{b})$$

And since $E_{\text{red}}^{\ominus} (\text{Ag}^+/\text{Ag}) > E_{\text{red}}^{\ominus} (\text{Ni}^{2+}/\text{Ni})$, we assign the (Ag^+/Ag) reaction to the cathode and the $(\text{Ni}^{2+}/\text{Ni})$ reaction to the anode. The reduction always takes place at the cathode

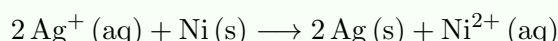


and therefore, Nickel is oxidized at the anode:



which was to be shown.

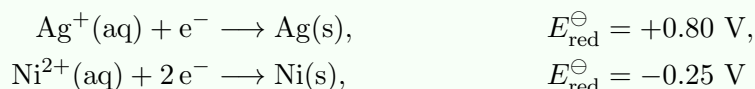
To see that they indeed produce the electrochemical reaction I, simply add reactions II and III together



which is the same as reaction I.

1.1b) Standard cell potential

The standard reduction potentials are



The standard cell potential is calculated from eq. (1), with silver reduced at the cathode and nickel oxidized at the anode. Therefore,

$$\begin{aligned} E_{\text{cell}}^{\ominus} &= E_{\text{red}}^{\ominus} (\text{Ag}^+/\text{Ag}) - E_{\text{red}}^{\ominus} (\text{Ni}^{2+}/\text{Ni}) \\ &= 0.80 \text{ V} - (-0.25 \text{ V}) \\ &= \boxed{1.05 \text{ V}} \end{aligned}$$

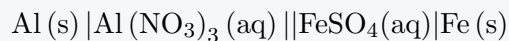
Note!: If $E_{\text{ox}}^{\ominus}$ is used, please remember to switch the sign:

$$E_{\text{ox}}^{\ominus} (\text{Ni}^{2+}/\text{Ni}) = -E_{\text{red}}^{\ominus} (\text{Ni}^{2+}/\text{Ni}) \implies E_{\text{cell}}^{\ominus} = E_{\text{red}}^{\ominus} (\text{Ag}^+/\text{Ag}) + E_{\text{ox}}^{\ominus} (\text{Ni}^{2+}/\text{Ni})$$

Both answers are equivalent.

Question: 1.2 Visualization of the galvanic cell

Draw a diagram of the following galvanic cell with a salt bridge of your choice



Solution: 1.2 Visualization of the galvanic cell

Figure 1 shows the diagram with KCl as the salt bridge of my choice.

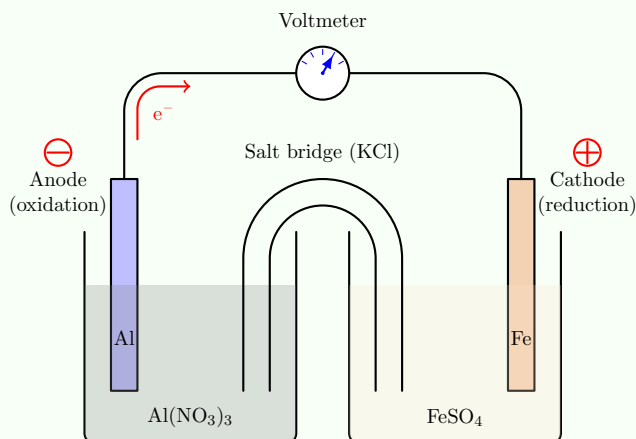


Figure 1: Diagram of the galvanic cell as denoted KCl salt bridge.

Question: 1.3 The cell

On slide 6, there is a diagram. For this diagram, please

- 1.3a) Write the cell designation.
- 1.3b) Write the half-cell and cell reaction(-s).
- 1.3c) Calculate the standard cell potential.

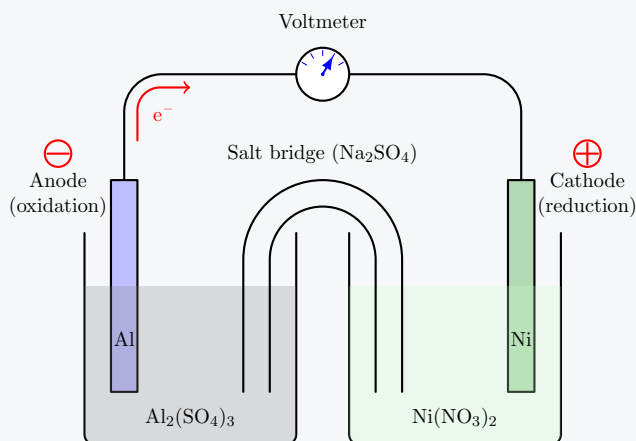
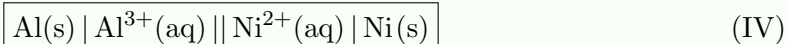


Figure 2: Reproduced Figure from slide 6.

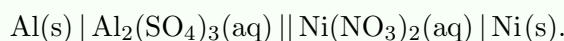
Solution: 1.3 The cell

1.3a) Cell designation

From Figure 2, aluminium is the anode and nickel is the cathode. Therefore, the cell designation is



Equivalently, using the salts shown in Figure 2, we can write



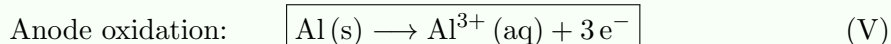
1.3b) Half-cell and cell reaction(-s)

From Table 1, we have

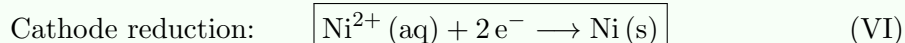
$$E_{\text{red}}^{\ominus}(\text{Al}^{3+}/\text{Al}) = -1.66 \text{ V} \quad (5\text{a})$$

$$E_{\text{red}}^{\ominus}(\text{Ni}^{2+}/\text{Ni}) = -0.25 \text{ V} \quad (5\text{b})$$

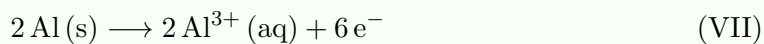
Aluminium is oxidized at the anode:



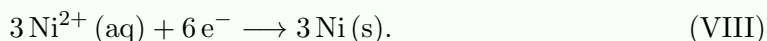
Nickel ions are reduced at the cathode:



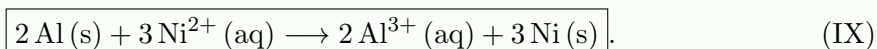
To obtain the cell reaction, the number of electrons must be equal in both half-cell reactions. Therefore, reaction V is multiplied by 2, and reaction VI is multiplied by 3:



and



Adding reactions VII and VIII gives the overall cell reaction



1.3c) Standard cell potential

The standard cell potential is calculated from eq. (1). Since nickel is reduced at the cathode and aluminium is oxidized at the anode, we obtain

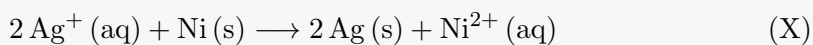
$$\begin{aligned} E_{\text{cell}}^{\ominus} &= E_{\text{red}}^{\ominus}(\text{Ni}^{2+}/\text{Ni}) - E_{\text{red}}^{\ominus}(\text{Al}^{3+}/\text{Al}) \\ &= -0.25 \text{ V} - (-1.66 \text{ V}) \\ &= \boxed{1.41 \text{ V}} \end{aligned}$$

Therefore, the standard cell potential of the cell shown in Figure 2 is 1.41 V.

2 The Cell Potential

Question: 2 The cell potential

2.1) For the following electrochemical reaction



2.1a) How many electrons participate in the half-cell reactions?

2.1b) Calculate the equilibrium constant of the reaction under standard conditions.

2.2) Consider the following cell

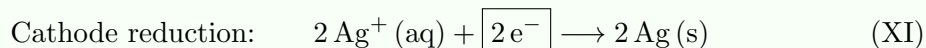


How much does the cell potential change when Q is increased by a factor 10?

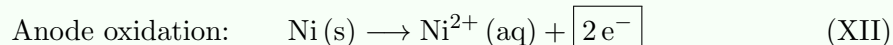
Solution: 2 The cell potential

2.1a) Number of electrons

From the solution of question 1.1, the half-cell reactions are



and



Therefore, the number of electrons participating in the half-cell reactions is

$$\boxed{\nu = 2}.$$

2.1b) Equilibrium constant under standard conditions

From question 1.1, the standard cell potential of the silver–nickel cell is

$$E_{\text{cell}}^{\ominus} = 1.05 \text{ V}$$

Using eq. (3), the equilibrium constant is

$$\begin{aligned} K &= \exp \left(\frac{\nu F E_{\text{cell}}^{\ominus}}{RT} \right) \\ &= \exp \left(\frac{2 \times 96485 \text{ C mol}^{-1} \times 1.05 \text{ V}}{8.314 \text{ J mol}^{-1} \text{ K}^{-1} \times 298.15 \text{ K}} \right) \\ &\approx \boxed{3.16 \times 10^{35}} \end{aligned}$$

which was to be shown.

2.2) Change in cell potential when Q is increased by a factor 10

The Nernst equation is

$$E_{\text{cell}} = E_{\text{cell}}^{\ominus} - \frac{RT}{\nu F} \ln(Q)$$

Let the reaction quotient be increased by a general factor a , i.e.

$$Q_{\text{new}} = aQ_{\text{old}}.$$

Then the new cell potential is

$$\begin{aligned} E_{\text{new}} &= E_{\text{cell}}^{\ominus} - \frac{RT}{\nu F} \ln(aQ_{\text{old}}) \\ &= E_{\text{cell}}^{\ominus} - \frac{RT}{\nu F} [\ln(a) + \ln(Q_{\text{old}})]. \end{aligned}$$

The old cell potential is

$$E_{\text{old}} = E_{\text{cell}}^{\ominus} - \frac{RT}{\nu F} \ln(Q_{\text{old}}).$$

Therefore, the change in cell potential is

$$\begin{aligned} \Delta E_{\text{cell}} &= E_{\text{new}} - E_{\text{old}} \\ &= -\frac{RT}{\nu F} \ln(a). \end{aligned}$$

Thus, the general result is

$$\Delta E_{\text{cell}} = -\frac{RT}{\nu F} \ln\left(\frac{Q_{\text{new}}}{Q_{\text{old}}}\right)$$

or, equivalently, if $Q_{\text{new}} = aQ_{\text{old}}$,

$$\Delta E_{\text{cell}} = -\frac{RT}{\nu F} \ln(a)$$

For the present problem, Q is increased by a factor 10, so

$$a = 10$$

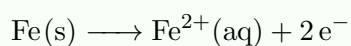
Therefore,

$$\Delta E_{\text{cell}} = -\frac{RT}{\nu F} \underbrace{\ln(10)}_{\approx 2.303}$$

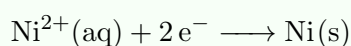
For the cell



the relevant half-cell reactions involve two electrons:



and



Thus,

$$\nu = 2.$$

At $T = 298.15$ K, we obtain

$$\begin{aligned}\Delta E_{\text{cell}} &= -\frac{8.314 \text{ J mol}^{-1} \text{ K}^{-1} \times 298.15 \text{ K}}{2 \times 96485 \text{ C mol}^{-1}} \ln(10) \\ &\approx -0.0296 \text{ V}.\end{aligned}$$

Therefore, when Q is increased by a factor 10, the cell potential changes by

$$\boxed{\Delta E_{\text{cell}} = -2.96 \times 10^{-2} \text{ V}}$$

or equivalently,

$$\boxed{\Delta E_{\text{cell}} = -29.6 \text{ mV}}.$$

Thus, the cell potential decreases by 29.6 mV.

References

- [1] O. Schmidt, “Exercise 1: Thermodynamics of electrochemical cells,” 2026. Problem sheet.